

SYNOPSIS

Report on

MUSECRAFTS : ECOMMERCE PLATFORM

by

Prasannajeet 202410116100145

Session:2025-2026 (IV Semester)

Under the supervision of

Ms. Shweta Singh

KIET Group of Institutions, Delhi-NCR, Ghaziabad



**DEPARTMENT OF COMPUTER APPLICATIONS
KIET GROUP OF INSTITUTIONS, DELHI-NCR,
GHAZIABAD-201206**

ABSTRACT

This project involves the design and development of the **MuseCraft E-Commerce Website**, a modern online shopping platform created to provide users with a seamless and engaging shopping experience. The primary objective was to build a responsive, user-friendly, and visually appealing website that enables customers to browse products, manage their carts, and place orders efficiently. The frontend was developed using **Next.js**, ensuring fast performance, smooth navigation, and compatibility across devices of varying screen sizes. The backend was built using **Node.js** and **Express.js**, while **MongoDB** was used for secure and scalable data management. The platform includes essential e-commerce features such as user authentication, product catalog management, shopping cart functionality, order processing, and an administrative dashboard for managing products, users, and orders. Special attention was given to UI/UX design to create an intuitive and engaging user experience. This project demonstrates the practical implementation of full-stack web development, database management, RESTful API integration, authentication, and deployment skills in a real-world e-commerce environment.

TABLE OF CONTENTS

	Page Number
1. Introduction	1-2
2. Literature Review	2
3. Project Objective	3
4. Hardware and Software Requirements	4
5. Project Flow	5-6
6. Project Outcome	7
7. Proposed Time Duration	8
References/ Bibliography	9

INTRODUCTION

In today's digital era, e-commerce platforms have transformed the way businesses and customers interact, making online shopping more convenient, efficient, and accessible than ever before. A well-designed e-commerce website not only enables customers to browse and purchase products seamlessly but also helps businesses manage inventory, orders, and customer interactions effectively through a centralized platform.

The project titled “**MuseCraft E-Commerce Website**” was undertaken with the objective of designing and developing a complete full-stack e-commerce solution that delivers a modern and engaging online shopping experience. The primary goal was to create a responsive, secure, and user-friendly platform where customers can explore products, add items to their cart, place orders, and manage their accounts with ease. Additionally, the project includes a dedicated admin panel that allows administrators to efficiently manage products, users, categories, and orders.

To achieve these objectives, the frontend was developed using **Next.js**, providing a fast, dynamic, and SEO-friendly user interface. The backend was built using **Node.js** and **Express.js**, enabling robust API development and business logic implementation. **MongoDB** was utilized as the database solution to ensure scalable and efficient data storage for products, users, and order information. Modern web development practices and UI/UX principles were adopted throughout the project to ensure smooth navigation, responsiveness, and compatibility across desktops, tablets, and mobile devices.

The platform incorporates essential e-commerce functionalities such as user authentication, product management, shopping cart operations, order processing, and administrative controls. Special attention was given to security through authentication and authorization mechanisms, ensuring a safe and reliable shopping environment for users. The development process focused

on creating a seamless experience for both customers and administrators while maintaining high performance and scalability.

This project not only highlights the technical aspects of full-stack web development but also demonstrates practical implementation of database management, RESTful API integration, authentication systems, and responsive interface design. By combining modern technologies with user-centered design principles, the MuseCraft E-Commerce Website serves as a comprehensive and professional solution for online retail management.

LITERATURE REVIEW

A literature review for a frontend development project such as the Ankuram Preschool, Muradnagar website examines existing research, best practices, and industry examples related to educational websites, UI/UX for parents, responsive design, accessibility, content strategy, and deployment. The goal is to situate the project within established knowledge, justify design choices, and identify features and approaches that improve usability, trust, and performance for preschool-focused audiences.

1. Educational Institutions & Digital Presence

Research and industry guidance consistently emphasize that schools and preschools require an online presence that communicates credibility, values, and practical information. A website for early-education institutions functions as both an information portal (admissions, fees, schedules) and a trust-building tool (staff profiles, photos, parent testimonials). Studies of school websites show parents prioritize clear contact information, program descriptions, and transparent admission procedures. Therefore, content clarity and hierarchy are central design concerns for preschool websites

2. User Experience (UX) for Parent/Guardian Audiences

Literature on UX for education-focused sites highlights that the primary user group—parents and guardians—seek quick access to essential information and reassurance about safety, curriculum, and staff competence. Key UX recommendations include:

- * **Clear Navigation:** Prioritized items (Admissions, Curriculum, About Us, Contact)
- * **Prominent Calls-to-Action:** (Visit, Enroll, Contact)
- * **Visual Storytelling:** (Photos of facilities, activities) to convey environment and ethos

* **Minimal Cognitive Load:** Concise text, readable typography, and consistent layout

These practices guided the project's emphasis on an intuitive layout and well-structured information architecture.

3. Responsive and Mobile-First Design

Mobile usage for search and information access has risen significantly; parents often browse on smartphones. The literature favors a mobile-first approach: design for the smallest screen first and progressively enhance for larger displays. Responsive web design (fluid grids, flexible images, media queries) ensures content adapts across devices, improving user retention and accessibility. Performance trade-offs on mobile (image sizes, script loading) are also well-documented; they influence decisions such as asset optimization and lazy loading in frontend implementations.

4. Accessibility and Inclusive Design

Accessibility research underlines the need to make educational websites usable for people with disabilities and diverse needs. Standards such as WCAG recommend semantic HTML, proper heading structure, alt text for images, sufficient color contrast, and keyboard navigability. For a preschool site, accessible content not only broadens reach but also reflects institutional responsibility. Implementing accessible components and ARIA attributes where necessary improves usability and compliance.

5. Visual Design & Branding for Early Education

Design literature specific to early-childhood education suggests a balance between playful and professional aesthetics. Visual cues—soft color palettes, rounded shapes, friendly imagery—evoke warmth and safety, while clean layouts and whitespace convey professionalism and clarity. Typography should be legible and welcoming; imagery should show engaged children and safe facilities. Consistent branding across pages helps reinforce identity and trust.

6. Performance, SEO, and Content Strategy

Fast-loading pages and clear metadata are critical for discoverability and user satisfaction. Best practices include compressing assets, minimizing render-blocking resources, and using semantic markup for better search engine indexing. Content strategy recommendations stress concise, parent-focused copy, FAQ sections, and visible contact pathways. Structured data (schema.org) for organization or local business can improve search visibility for local parents searching for preschools.

7. Deployment, Hosting, and Security

Modern frontend deployment practices advocate for static-site or Jamstack architectures when dynamic server-side logic is minimal. Platforms like Vercel offer continuous deployment, CDN-backed delivery, and easy custom-domain configuration. Literature and industry blogs recommend HTTPS by default, secure configuration of forms (validation, anti-spam), and privacy considerations for handling any parent or child data. Even when personal data is not stored, cookie/privacy notices and basic data-protection awareness are advisable.

8. Case Studies & Comparable Implementations

Analyses of existing preschool and small school websites reveal common successful patterns:

Homepage Elements for Ankuram Preschool

Ankuram Preschool should incorporate the following structural elements on its homepage to enhance user experience and engagement:

* **Clear Hero Section:** A prominent section showcasing the preschool's offerings and key features.

- * **Quick Links to Admissions:** Convenient links to facilitate the admissions process.
- * **Dedicated Pages:** Separate pages dedicated to curriculum, staff, gallery, and contact information.
- * **Testimonial or News Sections:** Display of testimonials or news articles to showcase the preschool's active community engagement.
- * **Integration of Maps and Contact Forms:** The inclusion of maps and contact forms to facilitate scheduling visits.

These structural elements, coupled with the use of visual proof such as photographs and virtual tours, will strengthen the Ankuram Preschool website's appeal and encourage visitors to explore further.

9. Trends & Emerging Practices

Recent literature points to a few emerging trends relevant to school websites:

- Micro-interactions and subtle animations enhance the perceived polish of the platform without compromising its performance.
- Virtual tours or short videos provide a comprehensive overview of classrooms and activities.
- Chat or inquiry widgets facilitate real-time parent inquiries, ensuring their privacy.
- Progressive enhancement ensures that essential information remains accessible even if advanced features encounter technical difficulties.

These trends inform optional enhancements that can be adopted over time.

Summary & Implications for the Project

The reviewed literature converges on several actionable conclusions that shaped the Ankuram Preschool frontend:

Key Objectives:

- * **Enhance Clarity and User Experience:** Prioritize the clarity of content and user navigation to ensure a seamless experience for parents.
- * **Mobile-First Responsiveness:** Implement a mobile-first responsive layout to optimize performance for devices with low-latency connections.
- * **Accessibility and Semantic Markup:** Utilize accessible and semantic HTML, adhering to Web Content Accessibility Guidelines (WCAG) recommendations.
- * **Professional Visuals and Trust:** Strive to balance a child-friendly visual style with professional structure to establish trustworthiness.
- * **Modern Platform Deployment:** Deploy the website on a modern platform (Vercel) to ensure fast and reliable delivery while enforcing HTTPS and fundamental security and privacy practices.

By aligning the project's design and technical decisions with these evidence-based practices and real-world examples, the Ankuram Preschool website aims to be a trustworthy, usable, and high-performing online presence that meets parental needs and supports institutional goals.

PROJECT OBJECTIVE

The primary objective of this project is to design, develop, and deploy a fully functional and user-friendly website for Ankuram Preschool, Muradnagar, that effectively represents the preschool's identity and provides parents and guardians with easy access to essential information. The specific objectives include:

1. Responsive Frontend Development:

Design a website layout that adapts seamlessly to various devices (desktop, tablet, and mobile) to guarantee accessibility and usability for all visitors.

2. Intuitive User Experience (UX) Design:

Create a clean and informative interface with user-friendly navigation, ensuring that parents and guardians can quickly access essential details such as admissions, curriculum, activities, and contact information.

3. Modern Web Technologies Implementation:

Utilize contemporary frontend frameworks and tools to achieve high performance, maintainability, and scalability of the website.

4. Stakeholder Collaboration:

Collaborate closely with the preschool's management team to align the website's design, content, and structure with their requirements and educational vision.

5. Professional Deployment:

Deploy the website on Vercel with a custom domain, managing hosting and configuration for optimal performance, high availability, and secure access.

6. Preschool Branding and Identity Integration:

Incorporate a child-friendly yet professional design aesthetic that communicates trust, safety, and nurturing values to parents.

7. Accessibility and Inclusivity Enhancement:

Adhere to web accessibility guidelines (WCAG) to ensure the website's usability for a broader range of users, including those with special needs.

8. Long-Term Sustainability:

Ensure that the codebase and deployment approach facilitate future updates, scalability, and seamless content modifications by the preschool team.

HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements

This project primarily focuses on frontend development and deployment, necessitating minimal hardware requirements to facilitate smooth development, testing, and deployment processes.

For Development Environment (Developer's System):

- * **Processor:** Intel i3 / AMD Ryzen 3 or equivalent
- * **RAM:** Minimum 4 GB (8 GB recommended for optimal multitasking)
- * **Storage:** 100 GB of free space (SSD is preferred for enhanced performance)
- * **Display:** 1366×768 resolution (Full HD is recommended)
- * **Internet Connection:** Stable broadband connection for seamless development, testing, and deployment

For End Users (Parents/Guardians visiting the website):

- * **Any device equipped with a web browser:** PC, Laptop, Tablet, Smartphone
- * **Internet Connectivity:** Minimum 2 Mbps for smooth browsing

Software Requirements

Development Tools & Frameworks:

- * **Frontend Framework/Library:** React.js / Next.js
- * **Markup & Styling:** HTML5, CSS3, Tailwind CSS
- * **JavaScript (ES6+):** For interactivity and logic implementation
- * **Version Control:** Git & GitHub for source code management
- * **Package Manager:** npm (Node Package Manager) or yarn

Development Environment:

- * **Operating System:** Windows / macOS / Linux (any modern operating system)
- * **Code Editor:** Visual Studio Code (preferred)
- * **Browser:** Google Chrome / Mozilla Firefox / Microsoft Edge (for testing and debugging)

Deployment & Hosting:

- * **Hosting Platform:** Vercel (for frontend deployment)
- * **Domain:** Custom domain configured with DNS settings

SSL Certificate: Provided by Vercel for secure HTTPS access.

Optional Tools:

- * Figma/Adobe XD (for UI/UX design and wireframing)
- * Postman (for testing APIs if backend integration is required in the future)

PROJECT FLOW

The development of the Ankuram Preschool Website followed a systematic process to ensure smooth execution and delivery. The project flow can be divided into the following key stages:

1. Requirements Gathering and Analysis

Initial meetings were held with the preschool management team to comprehend their expectations, target audience, and essential features.

Key requirements identified included:

- Admissions details
- Curriculum information
- Gallery
- Contact form
- Responsive design suitable for parents

2. Planning and Design

A sitemap and wireframes were created to define the website's structure. UI/UX design was prepared using tools like Figma, prioritizing simplicity, readability, and a child-friendly aesthetic. Feedback was collected from stakeholders prior to development.

3. Frontend Development

The website was developed using React.js/Next.js, HTML5, CSS3, and Tailwind CSS. Responsive design principles were applied to ensure compatibility across devices (desktop,

tablet, mobile). Navigation menus, content sections, and image galleries were coded with attention to performance and accessibility.

4. Integration and Testing

Cross-browser testing was conducted to verify compatibility with Chrome, Firefox, and Edge. Responsiveness was tested on various screen sizes and devices. Functionality such as navigation, forms, and image loading was thoroughly checked. Stakeholder review was conducted, and necessary refinements were applied.

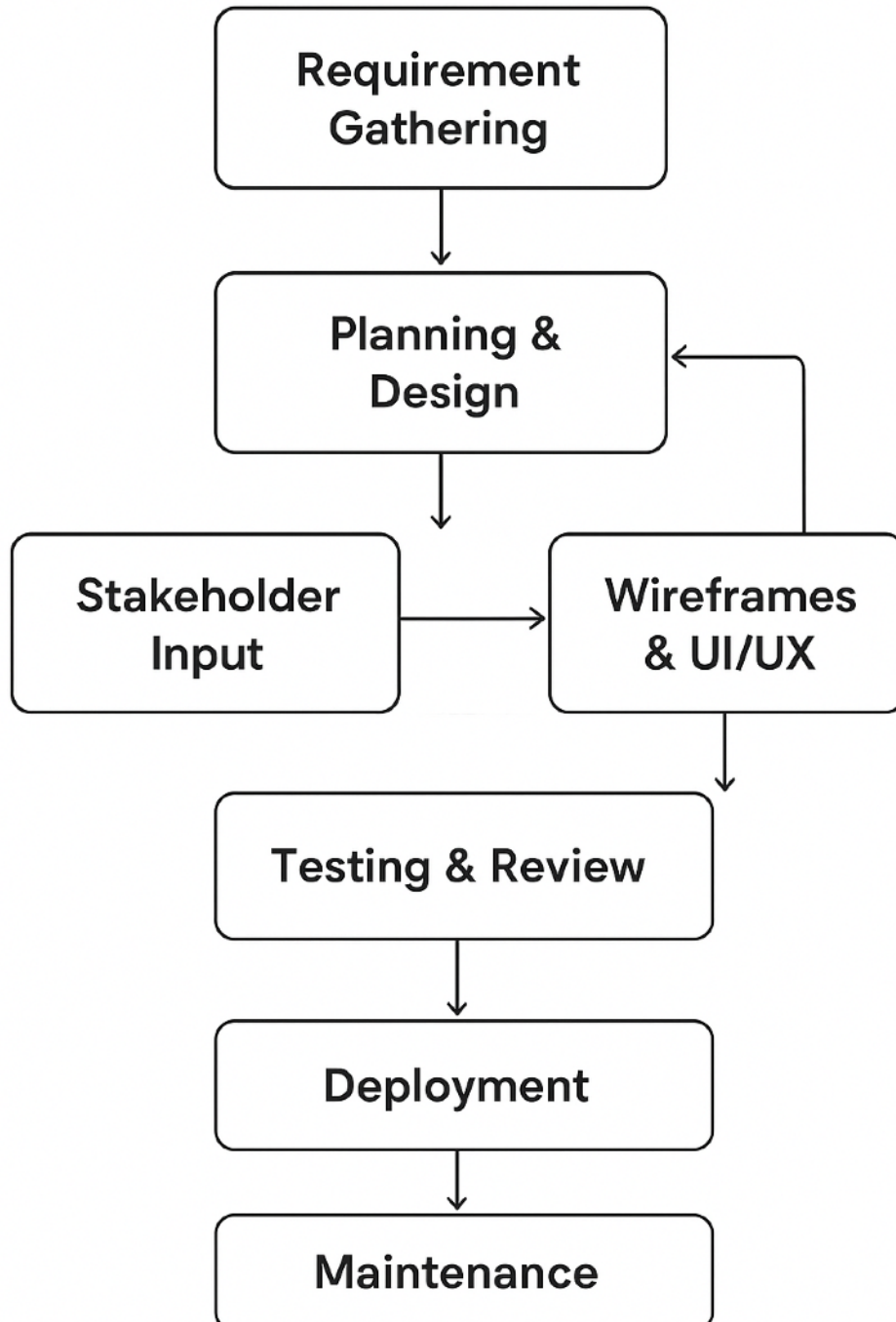
5. Deployment

The website was deployed on Vercel for fast and reliable hosting. A custom domain was configured for professional access, and SSL certificates were applied to ensure secure HTTPS browsing.

6. Maintenance and Future Enhancements

Basic documentation was provided to the preschool team for future content updates. Scope for future improvements includes adding features such as online admission forms, event updates, and parent feedback modules.

PROJECT FLOW



PROJECT OUTCOME

The development of the Ankuram Preschool, Muradnagar website successfully achieved its intended objectives, resulting in a fully functional, responsive, and user-friendly online platform. The outcomes of the project are as follows:

1.Functional Website Deployment:

A live, production-ready website was developed and deployed on Vercel with a custom domain, ensuring professional accessibility and secure HTTPS browsing.

2. Enhanced Online Presence:

The preschool now has a digital identity that communicates its vision, values, and facilities to parents and guardians effectively.

3. Responsive and Cross-Platform Design:

The website is optimized for various devices including desktops, tablets, and smartphones, providing a seamless user experience across platforms.

4.Improved User Experience (UX):

Clean navigation, intuitive layout, and structured content allow parents to easily find information related to admissions, curriculum, and contact details.

5. Child-Friendly Yet Professional Aesthetics:

The design balances playful visuals with a professional look, making it suitable for the preschool context while instilling trust among parents.

6. Stakeholder Satisfaction:

Continuous collaboration and feedback ensured that the final product matched the preschool management's expectations and requirements.

7. Scalability and Maintainability:

The project was developed using modern web technologies, making the website easily maintainable and capable of future expansion (such as adding online admission forms, event updates, or gallery enhancements).

8. Practical Learning Experience:

From a development perspective, this live project provided hands-on experience in frontend design, responsive development, deployment, and stakeholder collaboration, bridging the gap between academic learning and real-world application.

PROPOSED TIME DURATION

- **Requirement Gathering & Analysis – 1 Week**

Conducted meetings with preschool management to understand requirements, target audience, and essential features. Finalized the project scope and identified key functionalities such as admissions, curriculum, gallery, and contact details.

- **Planning & Design – 1 Week**

Designed the sitemap and wireframes for the website. Created UI/UX prototypes using design tools, focusing on a child-friendly yet professional look. Shared designs with stakeholders for feedback and approval.

- **Frontend Development – 2 Weeks**

Developed the website using React.js/Next.js, HTML5, CSS3, and Tailwind CSS. Implemented responsive layouts for different devices. Integrated preschool content such as text, images, and gallery sections. Ensured clean code and maintainable structure.

- **Testing & Refinement – 1 Week**

Performed cross-browser testing on Chrome, Firefox, and Edge. Checked responsiveness on multiple screen sizes (desktop, tablet, mobile). Debugged issues, optimized performance, and incorporated stakeholder feedback for improvements.

- **Deployment & Configuration – 1 Week**

Deployed the website on Vercel for hosting. Configured a custom domain with SSL certificate for secure HTTPS browsing. Conducted final verification of the live website to ensure smooth performance.

- **Documentation & Handover – 1 Week**

Prepared detailed project documentation covering design, development, and deployment processes. Provided the preschool team with basic update guidelines for content management. Officially handed over the live website for institutional use.

Total Duration: 7 Weeks (Approx. 2 Months)

REFERENCES

- 1. Mozilla Developer Network (MDN):** HTML, CSS, and JavaScript documentation. Available at: <https://developer.mozilla.org/>
- 2. React.js Official Documentation:** A JavaScript library for building user interfaces. Available at: <https://react.dev/>
- 3. Next.js Documentation:** The React framework for production. Available at: <https://nextjs.org/docs>
- 4. Tailwind CSS Documentation:** A utility-first CSS framework. Available at: <https://tailwindcss.com/docs>
- 5. Vercel Documentation:** A deployment and hosting platform for frontend projects. Available at: <https://vercel.com/docs>